UJJAWAL KUMAR

(Immediate Joiner)

Looking for Principal Data Scientist / Manager data science

Bengaluru, India

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English

EDUCATION

Bachelor of Engineering

New Horizon College 2010-2014

TECHNICAL SKILLS

SQL, Python, Machine Learning, Deep Learning, Gen AI, LLM, Lang graph, Agentic AI, AWS, Databricks, PySpark, TensorFlow, PyTorch, Power-BI

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Responsibilities

AI & Data Science Strategy

- **Directed** the Data Science and AI strategy across core business functions including **digital marketing, promotions, advertising, recommendation systems, demand forecasting, pricing, and Generative AI**, leveraging big data tools on cloud infrastructure.
- I play a **hands-on role** as well **led a team of Data Scientists**, driving innovation through task delegation, mentoring, and performance monitoring, while staying ahead of advancements in AI.
- **Developed and aligned** sophisticated AI-driven business strategies (e.g., Omni-channel analytics, recommendation engines) with executive objectives, ensuring a clear path to **revenue growth and operational efficiency**
- **Engaged with key stakeholders** and executive leadership, effectively translating complex data insights and model outputs into clear, **actionable business strategies** and investment decisions

ML Engineering & MLOps

- Led the **End-to-End ML Development lifecycle**, including optimizing system architecture and implementing scalable, data-driven processes that enhanced workflow efficiency
- **Established robust MLOps practices** by managing the full model lifecycle, utilizing **MLflow** for precise experiment tracking, version control, and continuous performance monitoring, ensuring governance and traceability at scale.
- **Standardized scalable ML model deployment** by leveraging **Docker** for packaging, implementing **Fast API-based microservices** for low-latency inference, and building interactive dashboards for real-time model visualization and stakeholder consumption

Projects and Industry 2014 – Present

Finance -> Deutsche Bank, ING Group & UBS

Retail -> Walmart, Amazon, Lowes

CPG - > AB InBev, General Mills

Automotive -> Toyota Motors, Mercedes

Pharma -> GSK, Boehringer Ingelheim

Transportation – Maersk, NS Netherlands, Deutsche Bahn

Projects and Industry

PWC Principal / Manager Data Science Aug 2025 – Oct 2025 (Impacted by Layoff)

- **Architected** and led the **deployment** of a **Multi-Agentic Generative AI solution** for personalized digital marketing content (text/image), integrating proprietary operational data via an **MCP Server** connection. This system achieved a 50% reduction in content creation time and drove a 6% lift in Conversion Rate (CVR).
  - Key Technologies: Multi-Agentic Frameworks (LangGraph/Autogen), LLMs (Claude/Llama/GPT/Gemini), Vector Databases (OpenSearch, Pinecone), MCP Server (Operational Data), Python, AWS Services (Sagemaker).

Publicis Groupe Sr /lead Data Scientist Apr 2022 – July 2025 (India & Netherlands)

- **Drove operational efficiency** for rail industry (**NS Netherlands and Deutsche Bahn**) by building a **Multi-Agentic RAG** architecture that securely connects to and processes live operational data (via **MCP server**) to offer instant, reliable information on train status and scheduling.
- **Architected** and led the production deployment of a **Multimodal, Multi-Agentic RAG system** for hyper-personalized marketing content (text, video, image generation), leveraging **Kubernetes and Docker** on Databricks to serve **millions of daily requests** with high reliability.
- Drove a **15% increase in ad campaign ROI** by building a real-time Dynamic Ad Generation Engine that customizes creative assets and copy based on granular customer behavior and engagement patterns.
- Architected and deployed a **Real-Time Dynamic Personalization Engine** leveraging **Reinforcement Learning** (RL) to determine the optimal next customer interaction, driving a 12% increase in Customer Lifetime Value (CLV).
- Minimized annual ad spend wastage by 6 million by designing and deploying an Advertising **Fraud Detection system** paired with a Dynamic Bid Adjustment model, improving overall campaign efficiency by 18%.
- Led the development and deployment of **Price Elasticity and Willingness-to-Pay** (WTP) Models, identifying and executing **optimal pricing strategies** that maximized quarterly revenue by 7% while maintaining customer purchase volume.
- **Developed and operationalized a Bayesian Media Mix Model (MMM) and Attribution framework** to precisely measure and optimize a multi-million-dollar marketing budget, resulting in a **10% reallocation of spend** to higher-performing channels.
- **Led a strategic initiative** to deliver **data-driven consumer insights** for CPG and Retail sectors, directly influencing product roadmaps and Go-to-Market strategies for **three major Amazon Private Label launches**.
  - **Engineered and productionized NLP models (e.g., Transformers)** for **Brand Perception and Sentiment Analysis**, converting thousands of open-ended survey responses into actionable product and marketing intelligence.
  - Defined **high-value market segments** and customer profiles using clustering and predictive modelling, enabling **tailored product positioning** that captured new market share

Publicis Groupe –TU Delft Netherlands Freelance Data Scientist 1st July 21– 15th Mar 22

- **Designed and developed AI-driven marketing analytics, recommendation, pricing & optimization models**, including:
  - Developed ML solutions on solar cells, identify defect in manufactured **solar cell material** and **forecasts material** performance under different conditions, which improvised solar energy production and efficient decision-making capabilities in energy auction markets
  - Price Optimization & **Forecasting** models to recommend real-time price adjustments based on market conditions and forecasted revenue and margin impact under different pricing strategies

Projects and Industry

Siemens Data Scientist 20th July 2020 – 03rd May 2021

- Led a data science team in developing **GAN based Anomaly Detection solution** using Deep Learning frameworks, enabling automated identification of anomalies across transactions, images, and other business datasets, improving operational efficiency and risk mitigation.
- Worked closely with the Product team of Siemens Mobility to develop **a News Analytics-driven Demand Prediction** system, leveraging NLP and time-series forecasting to enhance supply chain responsiveness across multiple sectors.

Infosys Data Scientist 8th Oct 2018 – 10th July 2020

- **Developed an AI-Driven Predictive Maintenance System** for Siemens, later adapted for ABB Power and General Electric, to enable fault detection in power grids, wind turbines, and substations. The solution reduced operational costs, improved quality assurance, and enhanced equipment longevity, driving efficiency in industrial energy systems.
  - **GAN**-powered **Defect Detection system**, designed for manufacturing quality control, identifying microscopic defects in critical components like turbine blades, circuit boards, and automotive parts.
  - Built a Predictive Maintenance System using **GAN-generated synthetic sensor data**, enhancing **fault detection** in industrial machinery. By simulating rare failure scenarios, the model improved predictive maintenance accuracy and reduced unplanned downtime for motors, turbines, and conveyor systems.
- **Demand Forecasting & Inventory Optimization for Retail**
  - Developed a multi-level demand forecasting solution using Time Series models (**LSTM**, Prophet, XGBoost) to predict sales demand across various SKUs.
  - Integrated supply chain optimization algorithms, reducing stockouts and overstock by 6%.
- Developed a Multi-Label Image Classification Tool for Toyota Motors Europe, enhancing image quality by denoising and converting 2D images to 3D for improved analysis and visualization.

Capgemini Data Scientist 19th Nov 2015 – 10th Sept 2018

- Partnered with Mercedes-Benz's Product Team to analyze user journey patterns using sequence embedding techniques to develop ML-based conversion prediction model, forecasting user actions and **next-best action recommendations**, enhancing customer engagement and conversion rates.
- Built an AI-Driven Customer Feedback & **Recommendation** System for Morgan Stanley, automatically tagging feedback to extract insights, improving customer experience, increasing adoption rates by 5%, identified new marketing strategies
- Developed **Forecasting & Inventory Optimization** Solutions for **CPG** in collaboration with Product Team.
  - Enabled risk reduction and data-driven financial decision-making, improving profit margins by 6–9% through optimized inventory planning.

Freelance Data Scientist 10th June 2014 – 10th Nov 2015

- Executed diverse high-impact data science and AI projects for various clients, leveraging deep expertise across the ML spectrum. This included **architecting and deploying robust time-series forecasting and classical ML** models for domains like sales, inventory, and customer churn, these projects spanned the full data lifecycle from leading comprehensive exploratory data analysis (**EDA**) and **segmentation** to delivering actionable strategic insights via automated performance **dashboards**.